Analysis group meeting

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6/2/25

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Cooking

- Cooking for the MC vs Data study:
  - Data (IBCuSn, with and without denoising) is done
  - MC (IBCuSn, 40%-60% , with and without denoising) is done

- Pass0v11 will be done with COATJAVA/12.0.5t
- Using the new network OBLD2 and denoising.

MC vs. Data study

- Compare MC and data for inbending CuSn:
  - Simulate 40%Cu - 60%Sn at 90, 95 and 100 nA (main currents for outbending data)
  - Use OB runs for the background
  - Data: runs 18348, 18354, 18372, 18373, 18394
  - Current 100,110,130,150, and 175.
  - Discrepancy between data and simulation currents, but I think I can't produce background files with enough stats from the IB runs.

    But I can try.

Outbending

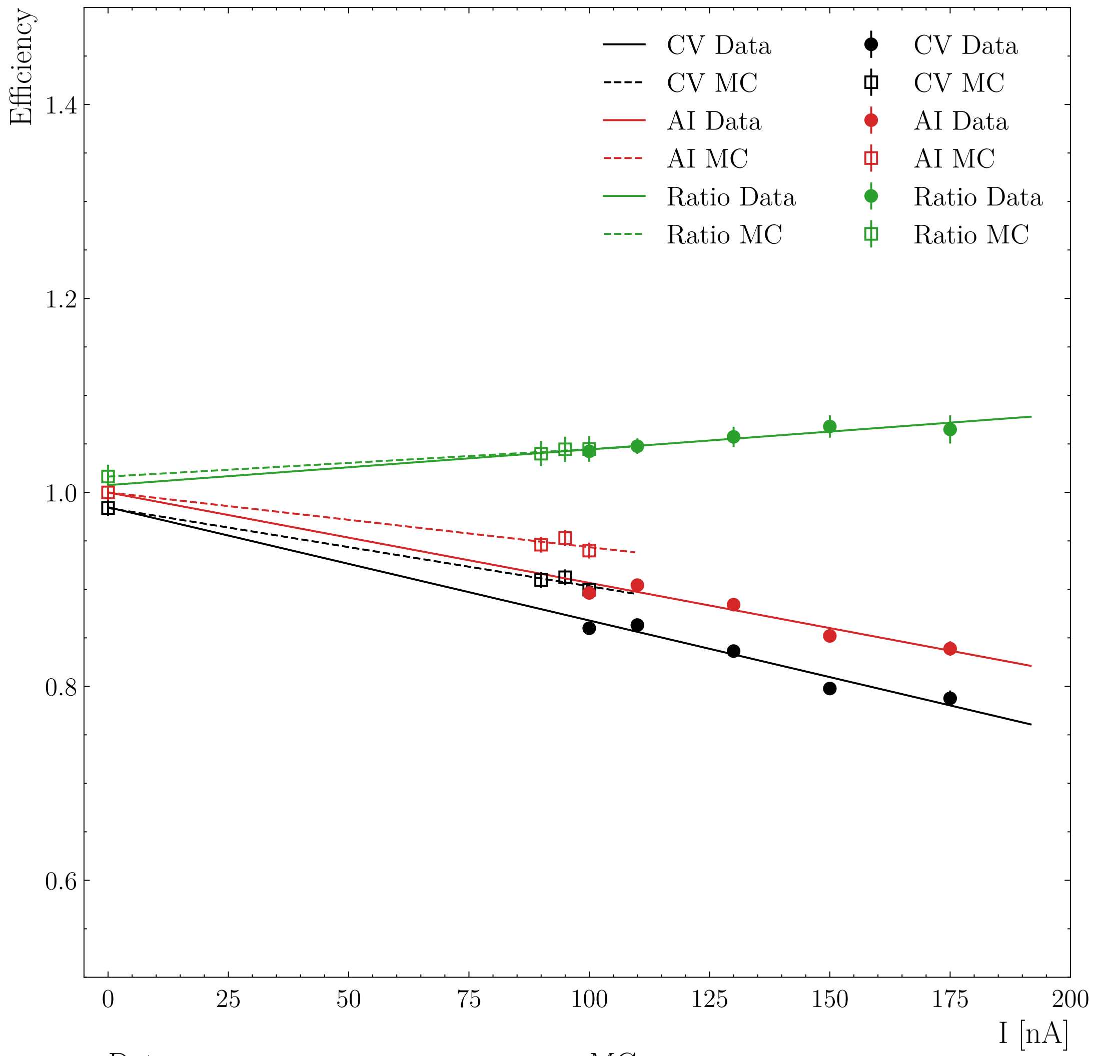
```
All the current: [20, 30, 85, 86, 90, 95, 100]
{20: {'# of Events': 30344699, 'Current (nA)': 20},
 30: {'# of Events': 39255714, 'Current (nA)': 30},
 85: {'# of Events': 11277806, 'Current (nA)': 85},
 86: {'# of Events': 231303974, 'Current (nA)': 86},
 90: {'# of Events': 3626488330, 'Current (nA)': 90},
 95: {'# of Events': 9827747103, 'Current (nA)': 95},
 100: {'# of Events': 4459965291, 'Current (nA)': 100}}
```

Inbending

```
All the current: [100, 110, 130, 135, 150, 175]
{100: {'# of Events': 16164206, 'Current (nA)': 100},
 110: {'# of Events': 39605751, 'Current (nA)': 110},
 130: {'# of Events': 498246511, 'Current (nA)': 130},
 135: {'# of Events': 22294711, 'Current (nA)': 135},
 150: {'# of Events': 475412781, 'Current (nA)': 150},
 175: {'# of Events': 26278056, 'Current (nA)': 175}}
```

MC vs. Data study

Comparison CuSn MC vs. Data with denoising using OBLD2 network



Data

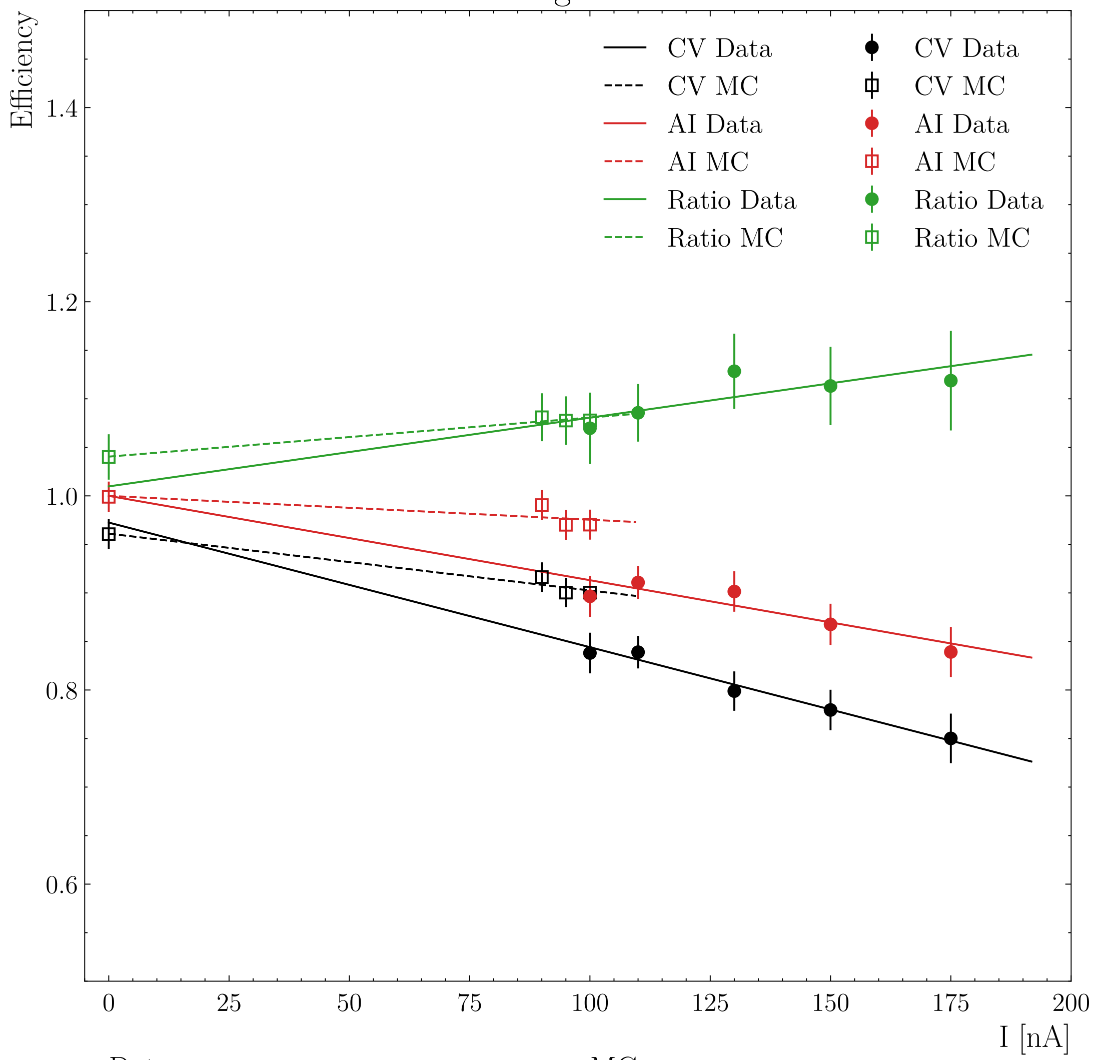
1.00000 + -0.000933 x (±0.000106)

0.98472 + -0.001168 x (±0.000112)

MC

1.00000 + -0.000565 x (±0.000101)

0.98395 + -0.000807 x (±0.000101)



Data

1.00000 + -0.000869 x (±0.000370)

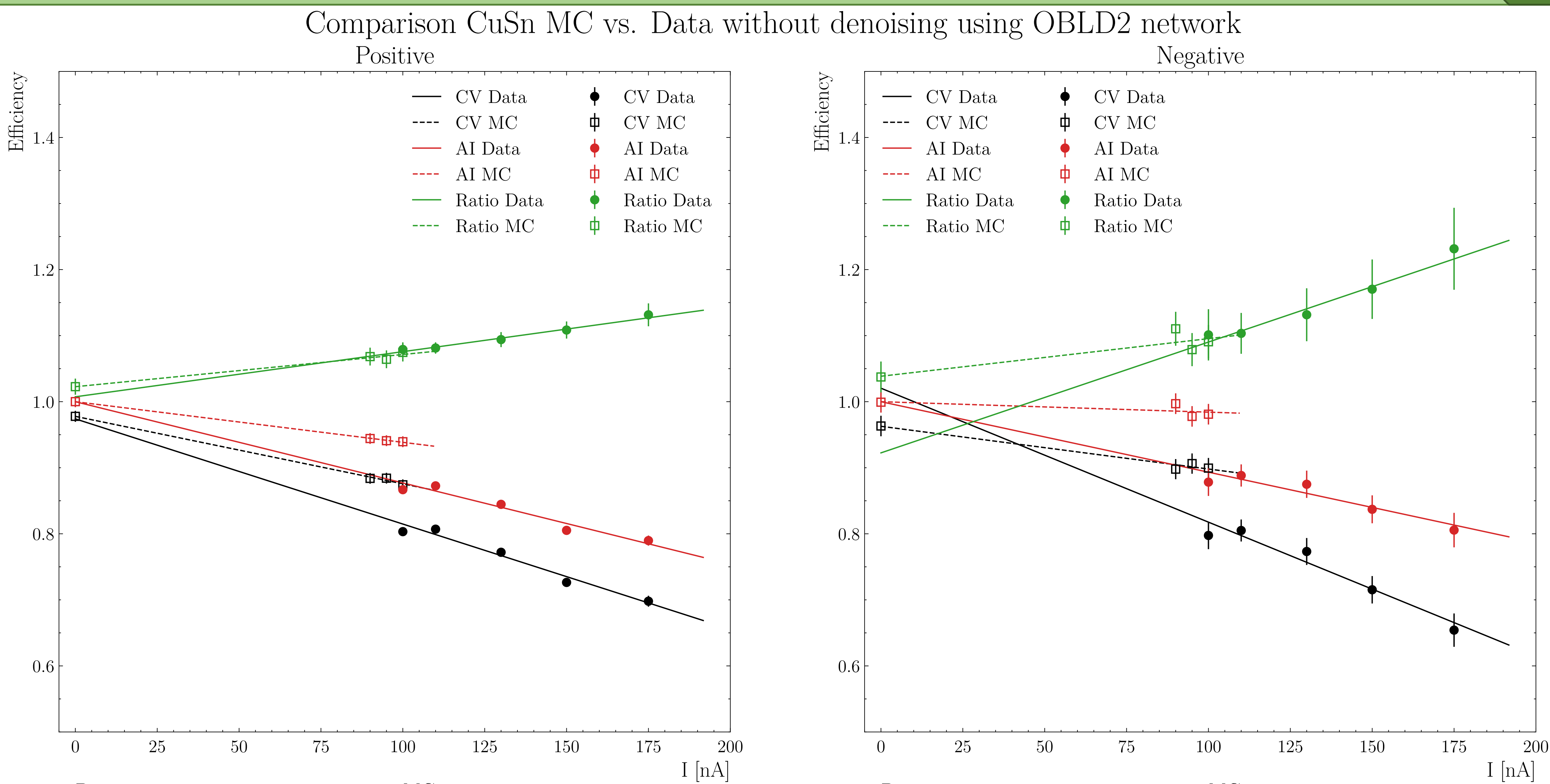
0.97249 + -0.001284 x (±0.000366)

MC

1.00000 + -0.000246 x (±0.000189)

0.96109 + -0.000586 x (±0.000186)

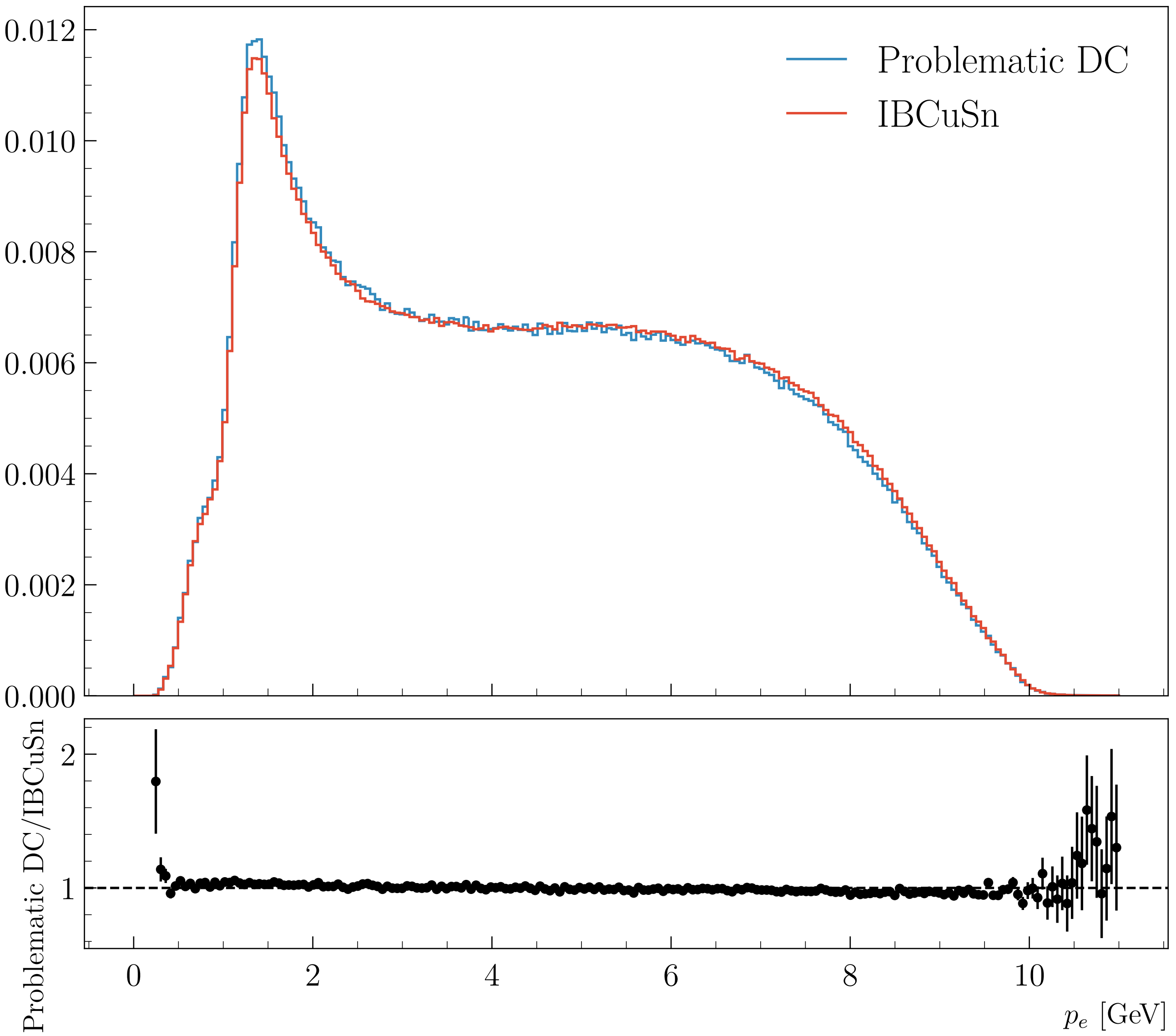
MC vs. Data study



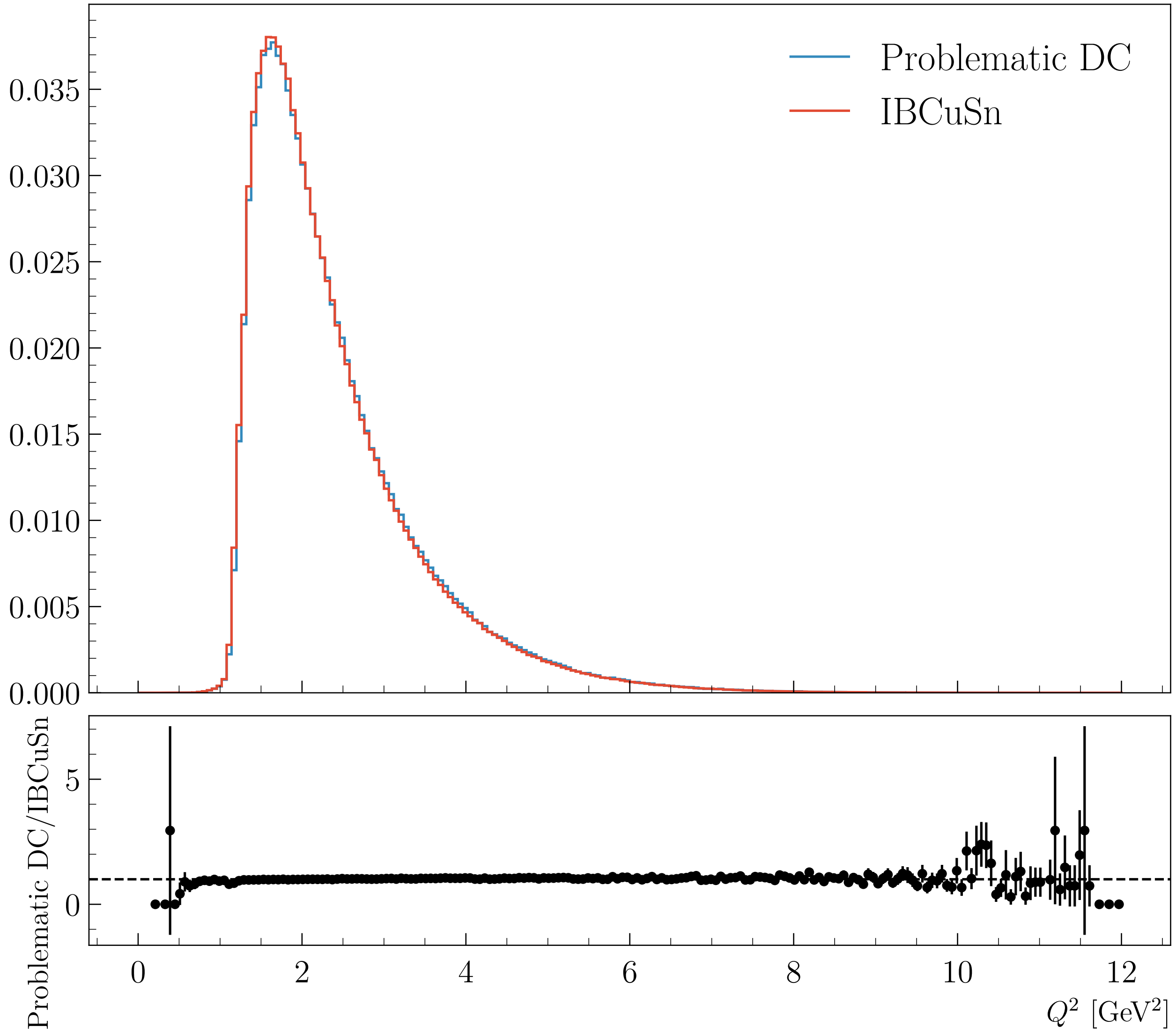
Data	MC	Data	MC
1.00000 + -0.001229 x (±0.000105)	1.00000 + -0.000615 x (±0.000101)	1.00000 + -0.001067 x (±0.000371)	1.00000 + -0.000158 x (±0.000190)
0.97399 + -0.001592 x (±0.000113)	0.97780 + -0.001021 x (±0.000100)	1.02041 + -0.002028 x (±0.000363)	0.96276 + -0.000647 x (±0.000188)

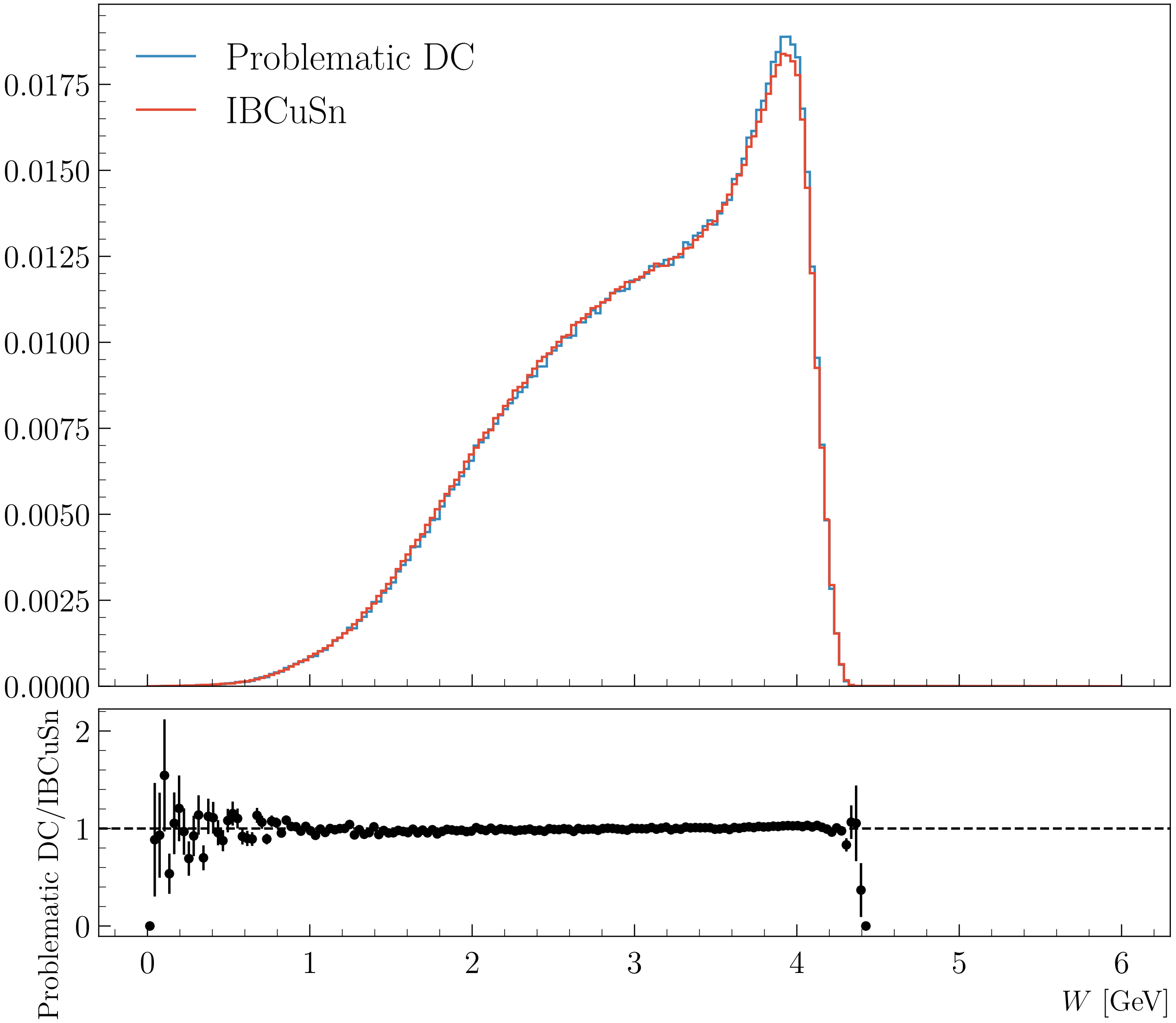
Physics quantities with problematics DC runs

- Runs: 18358, 18359, 18360, 18361, 18362, 18363, 18364, 18365, 18366, 18368
- Electron: pid = 11, status < 0. and -9.217365 < Vz < 5 cm



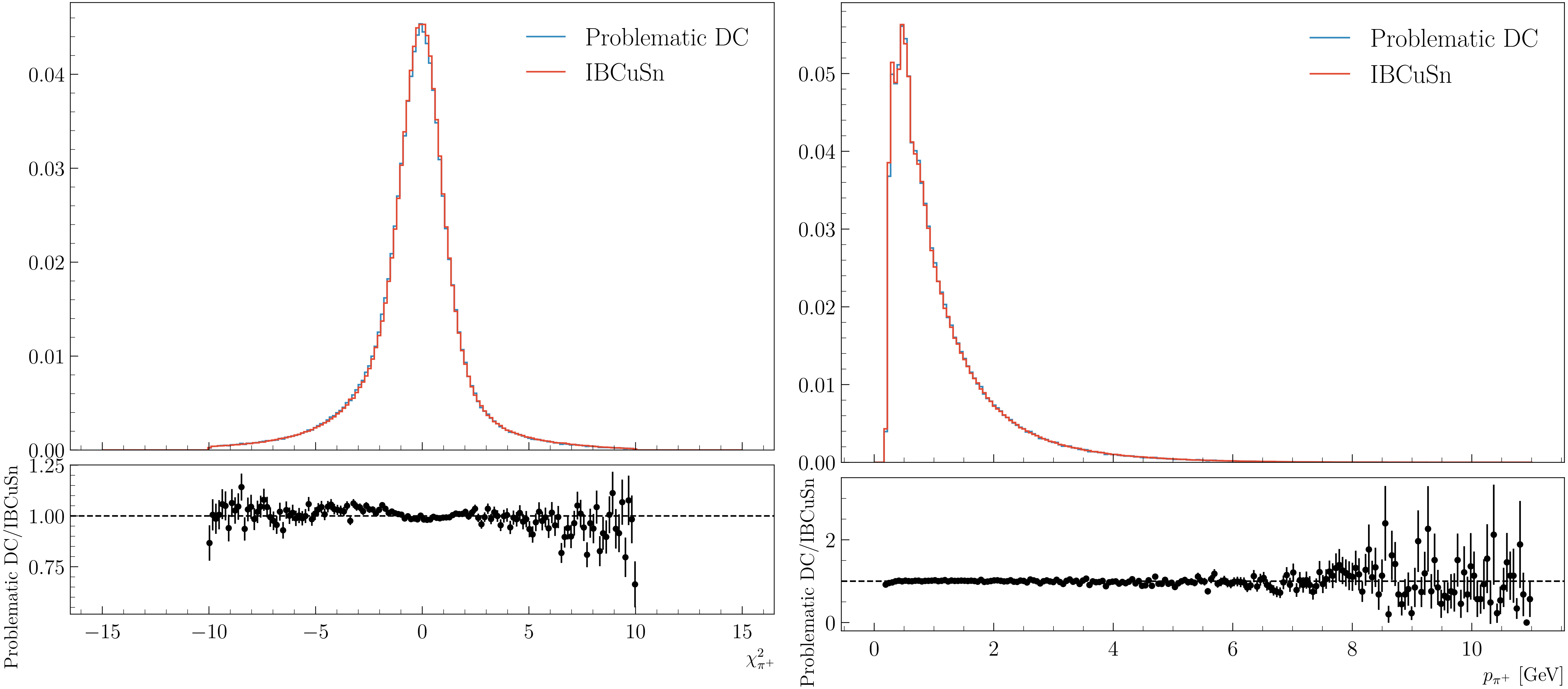
Physics quantities with problematics DC runs



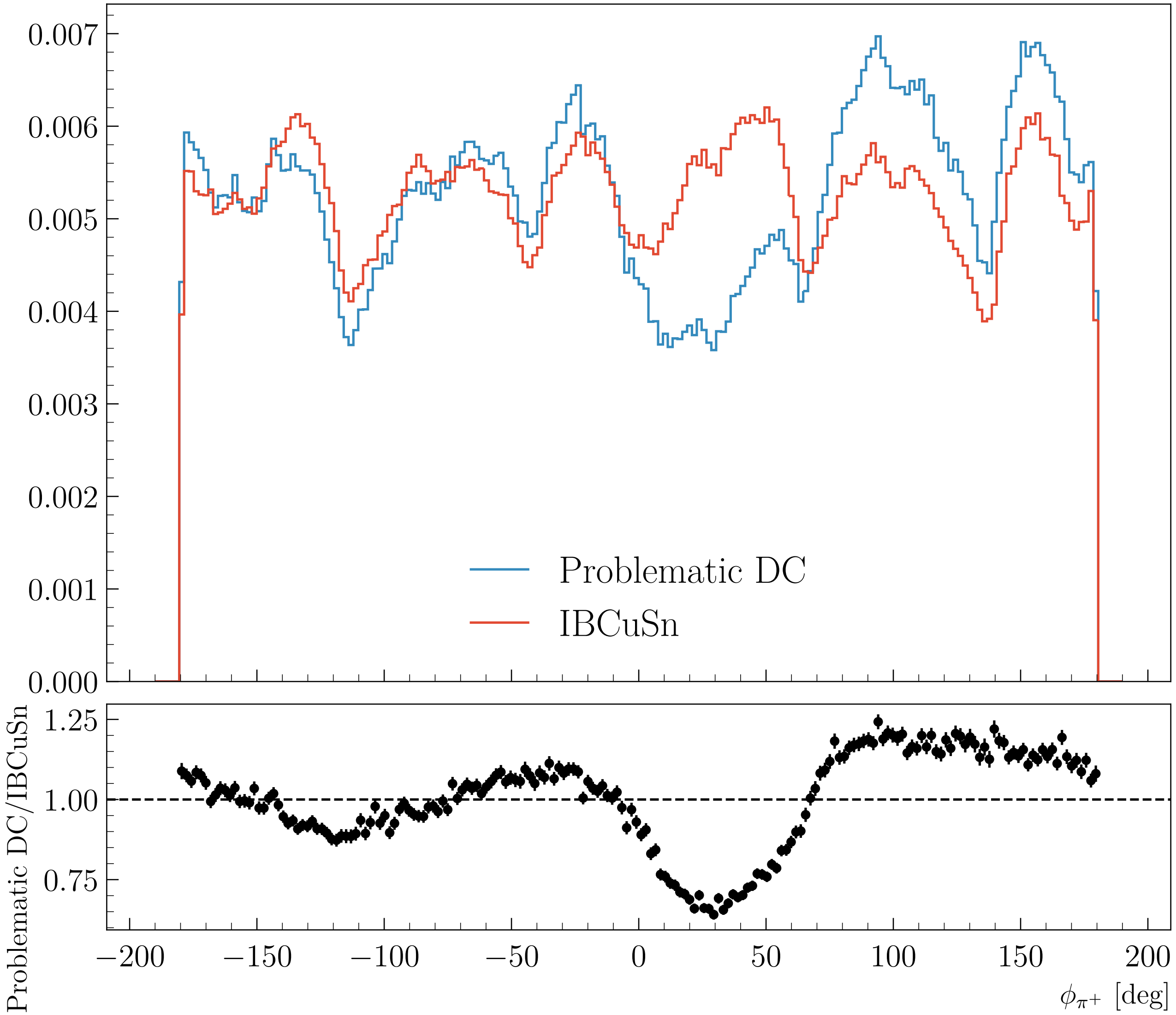


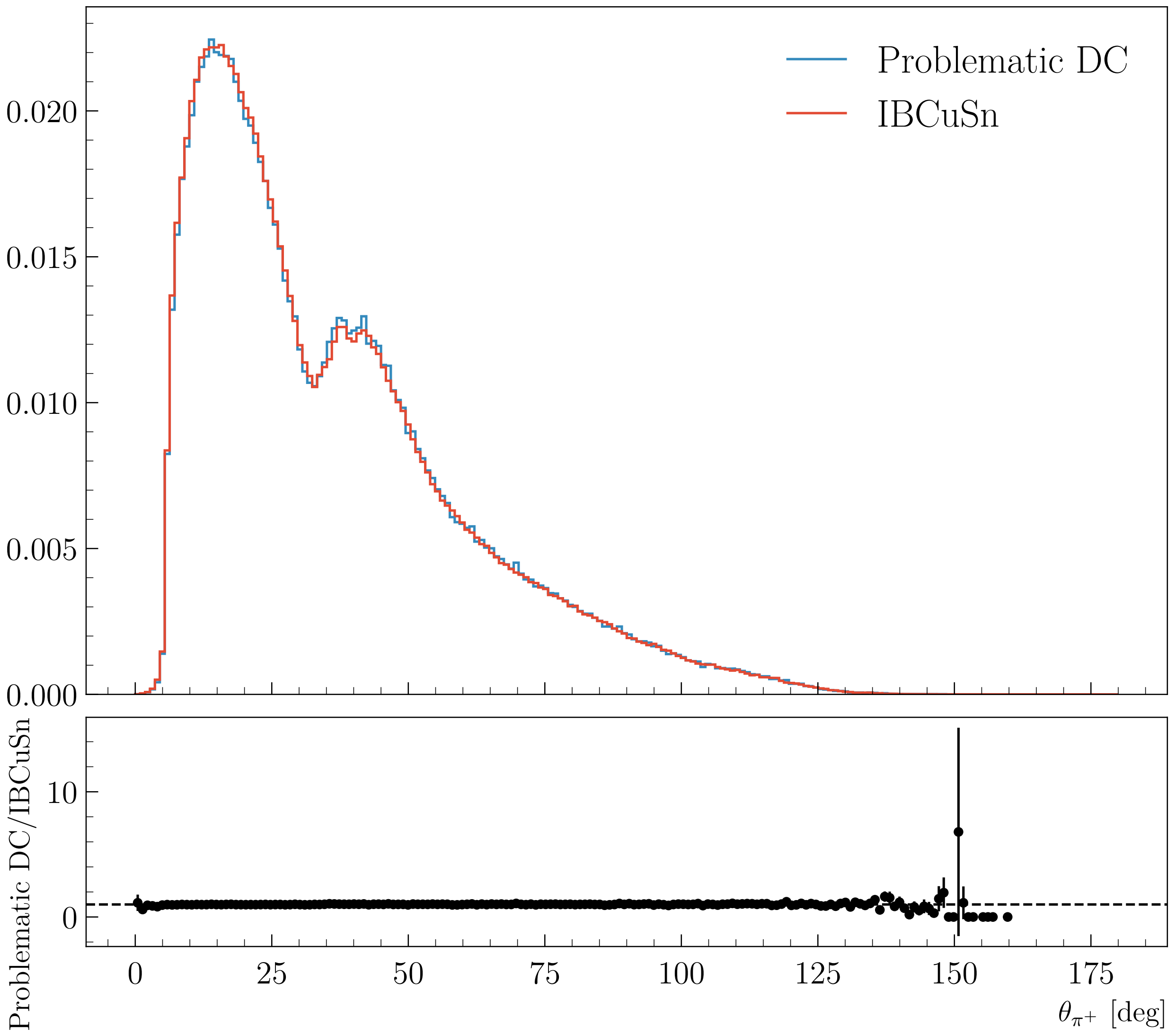
Physics quantities with problematics DC runs

Cut pi+: -10 < chi2pid <10



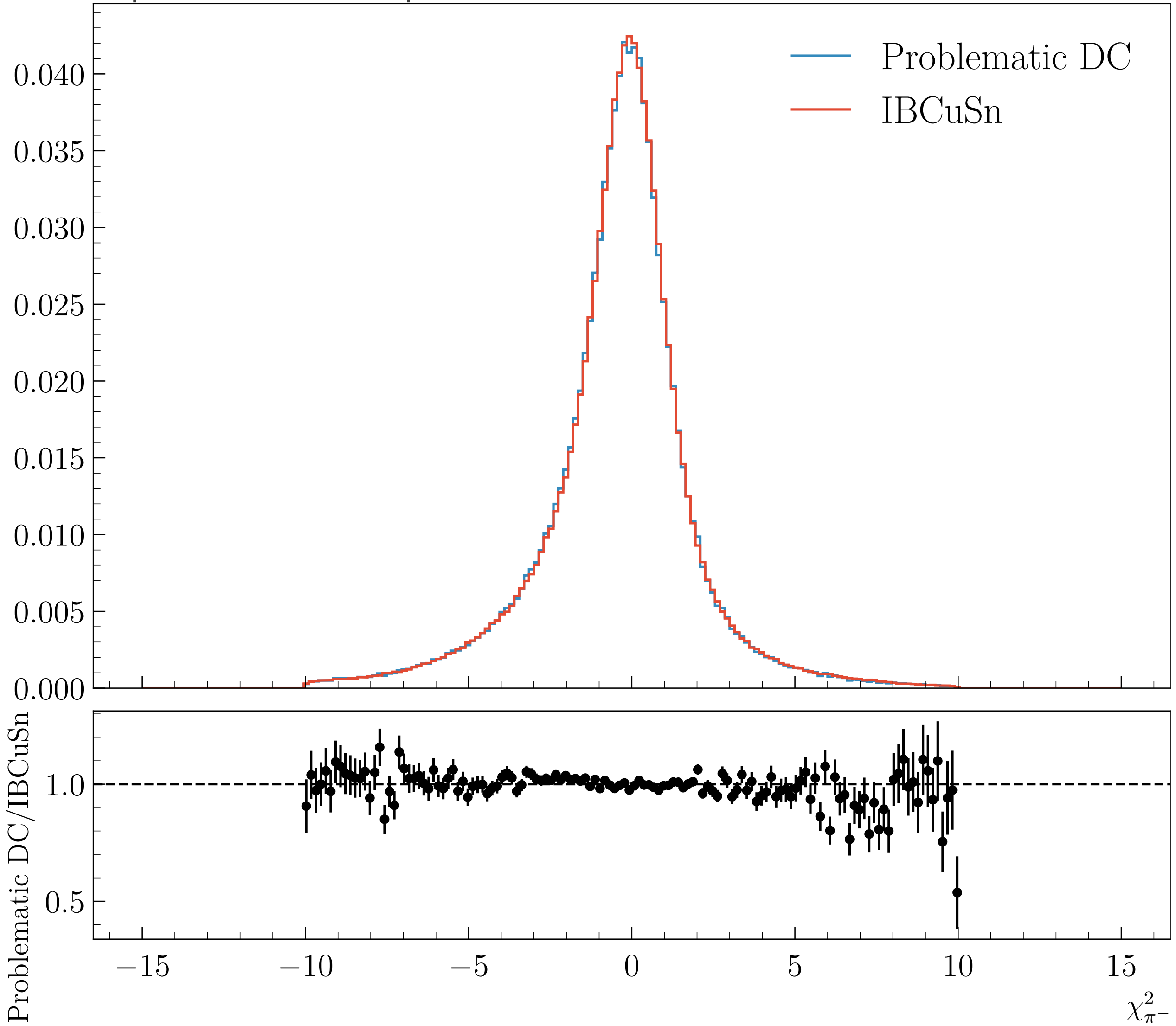
Physics quantities with problematics DC runs

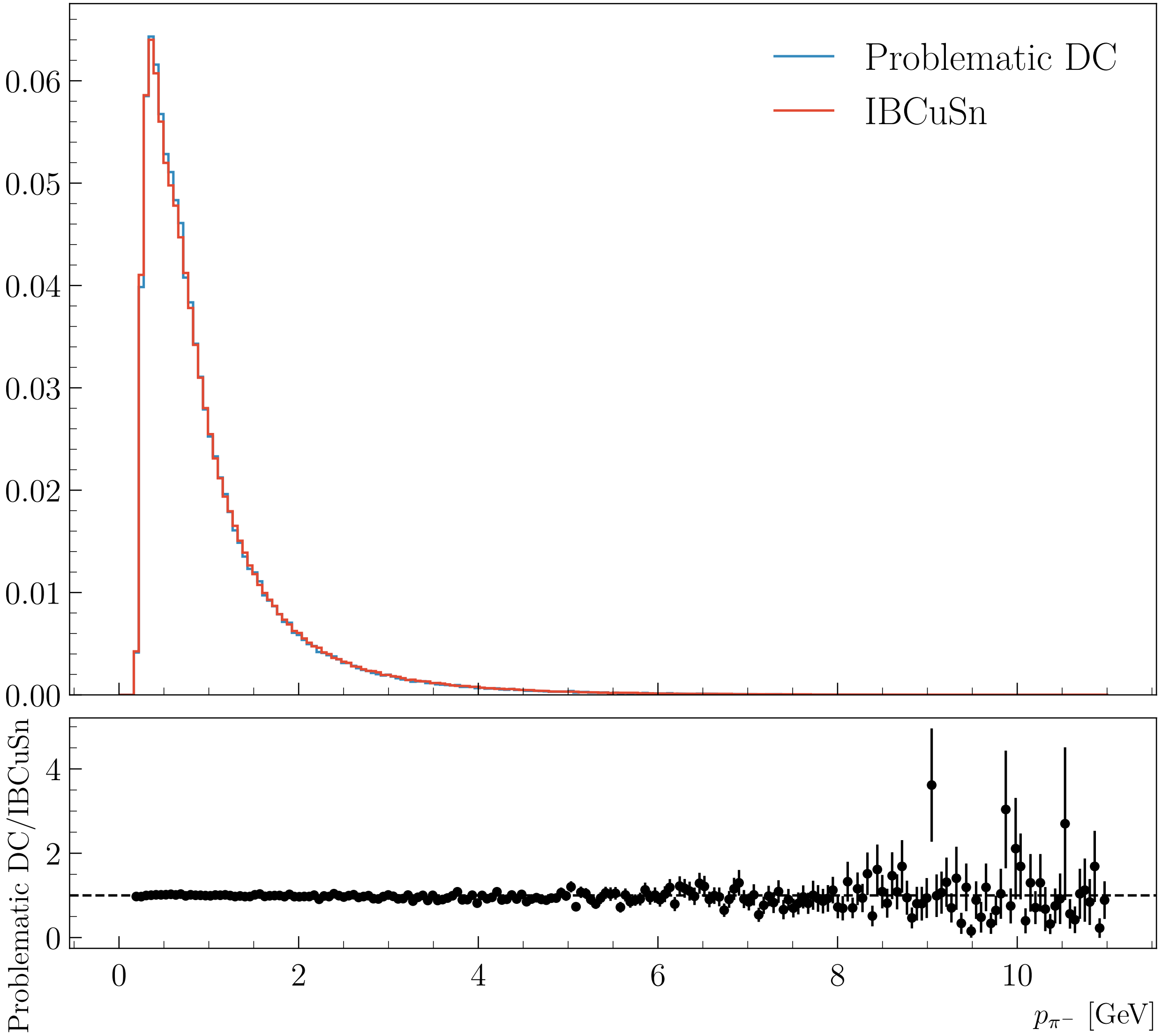


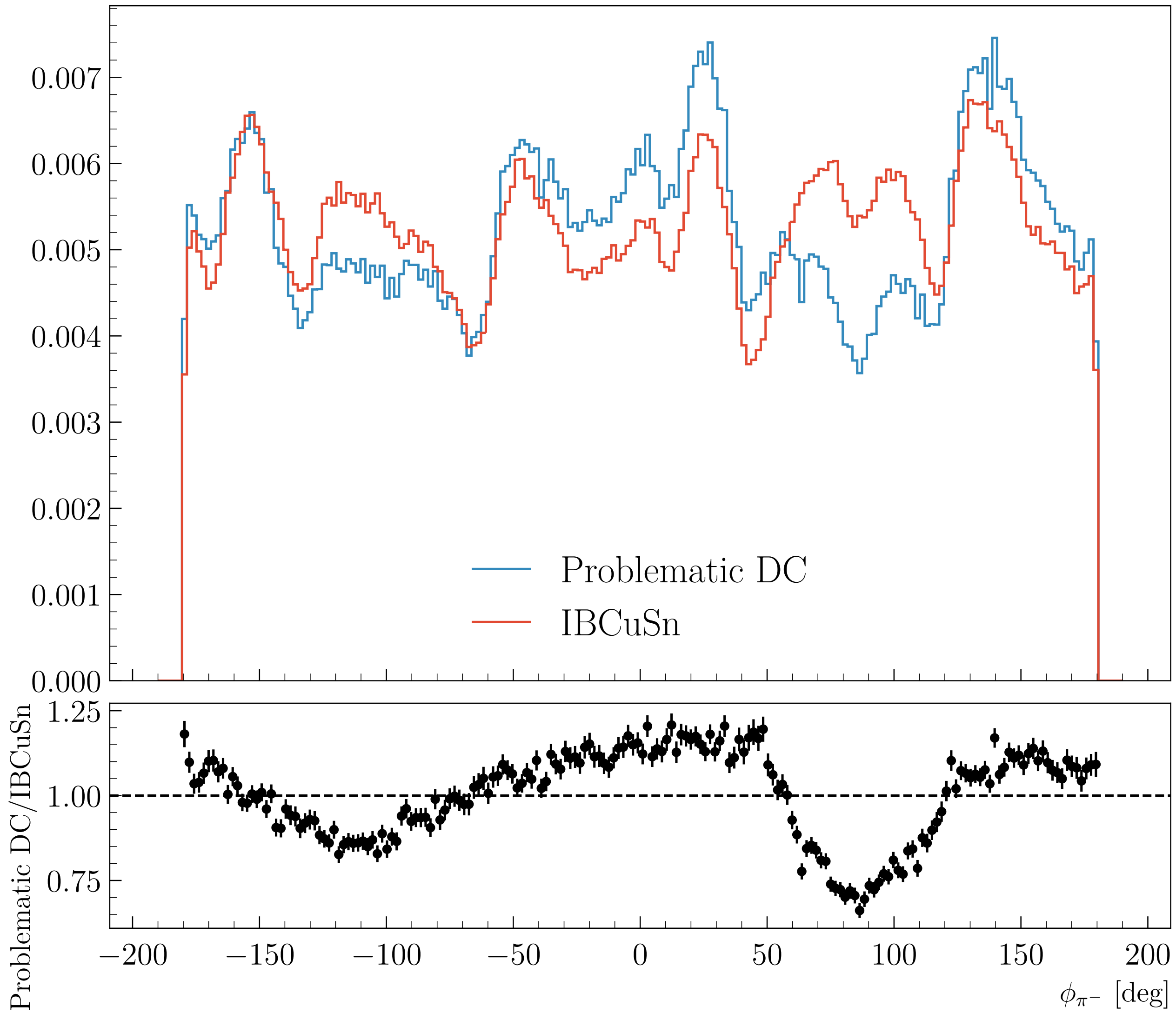


Physics quantities with problematics DC runs

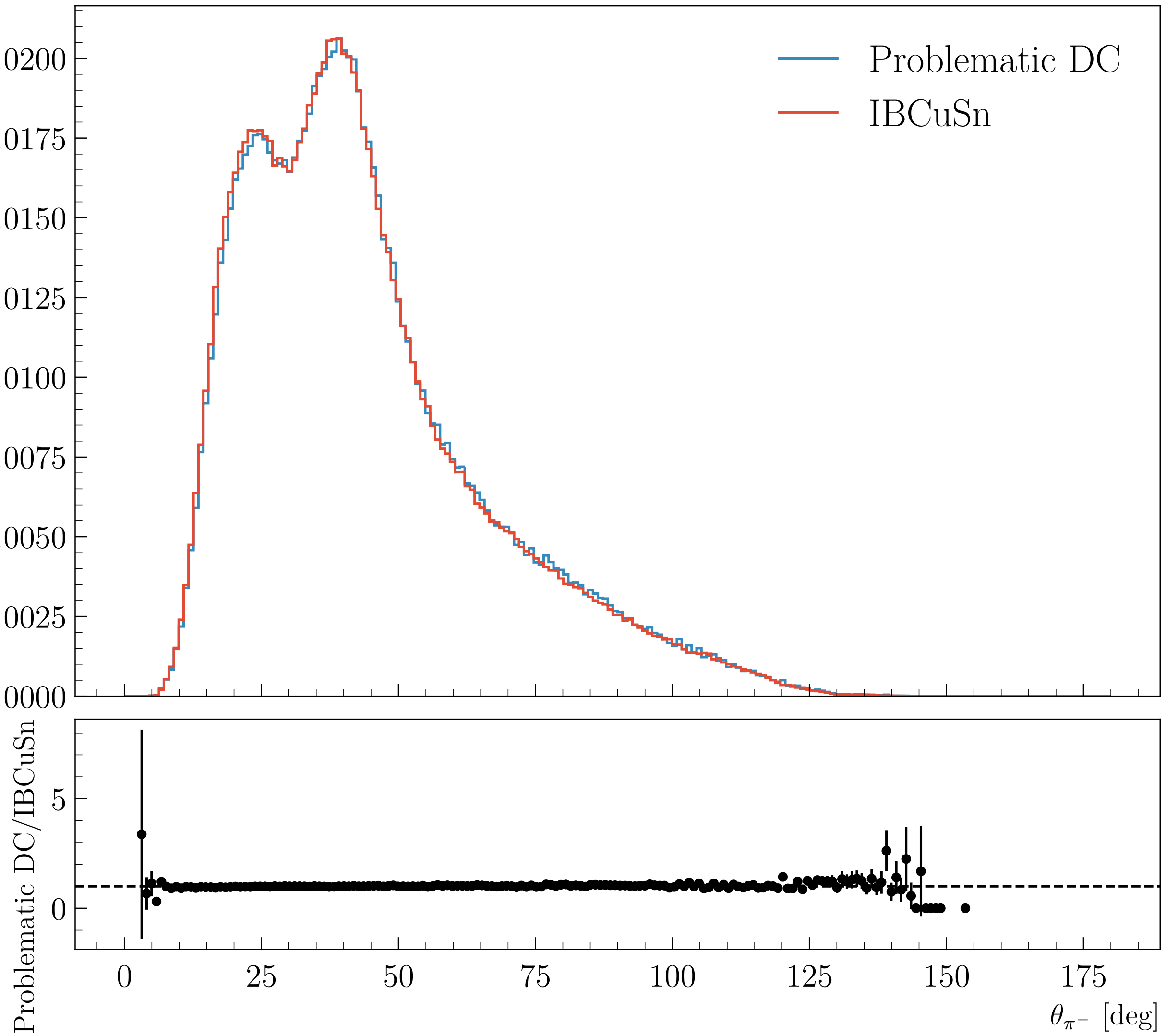
Cut pi-: -10 < chi2pid <10





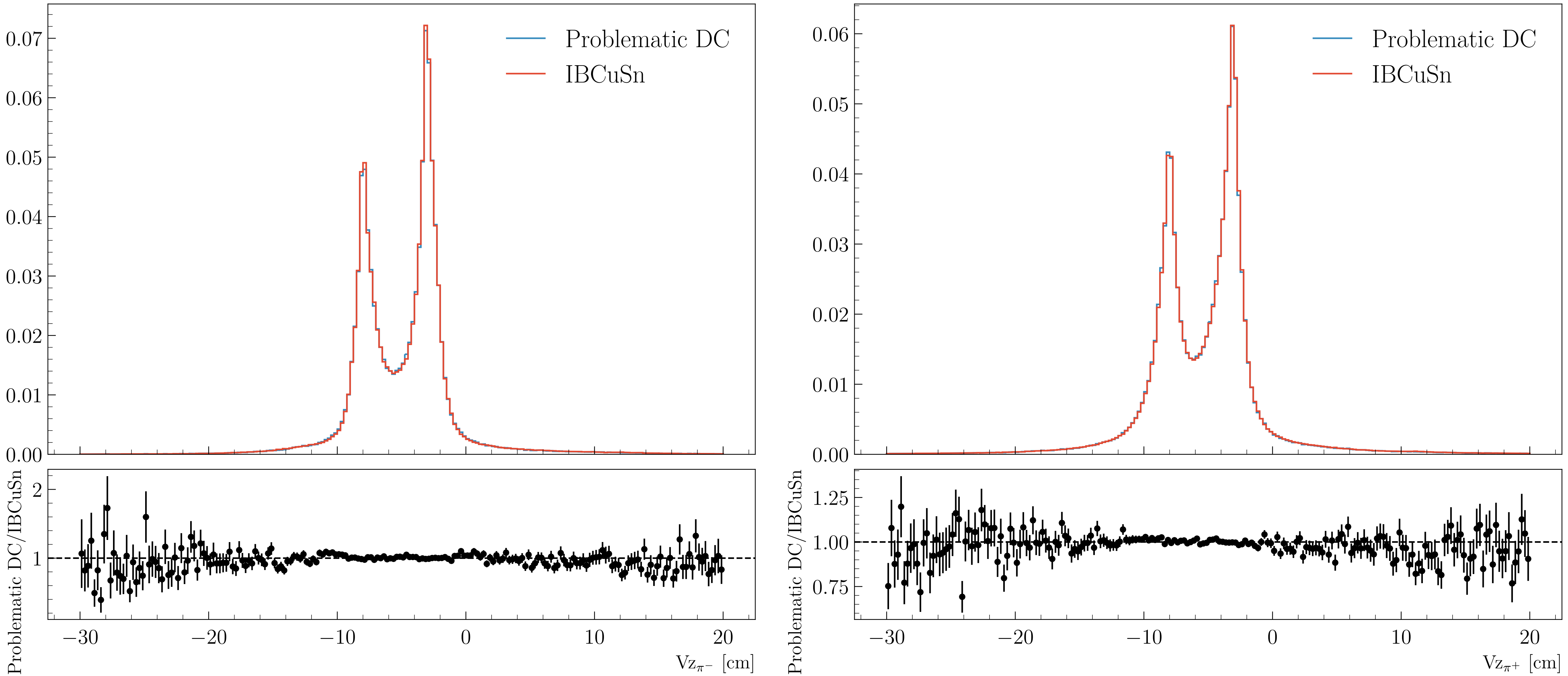


0.007
0.006
0.005
0.004
0.003
0.002
0.001
0.000
Problematic DC
IBCuSn
1.25
1.00
0.75
Problematic DC/IBCuSn
−200
−150
−100
−50
0
50
100
150
200
ϕ_{π^-} [deg]

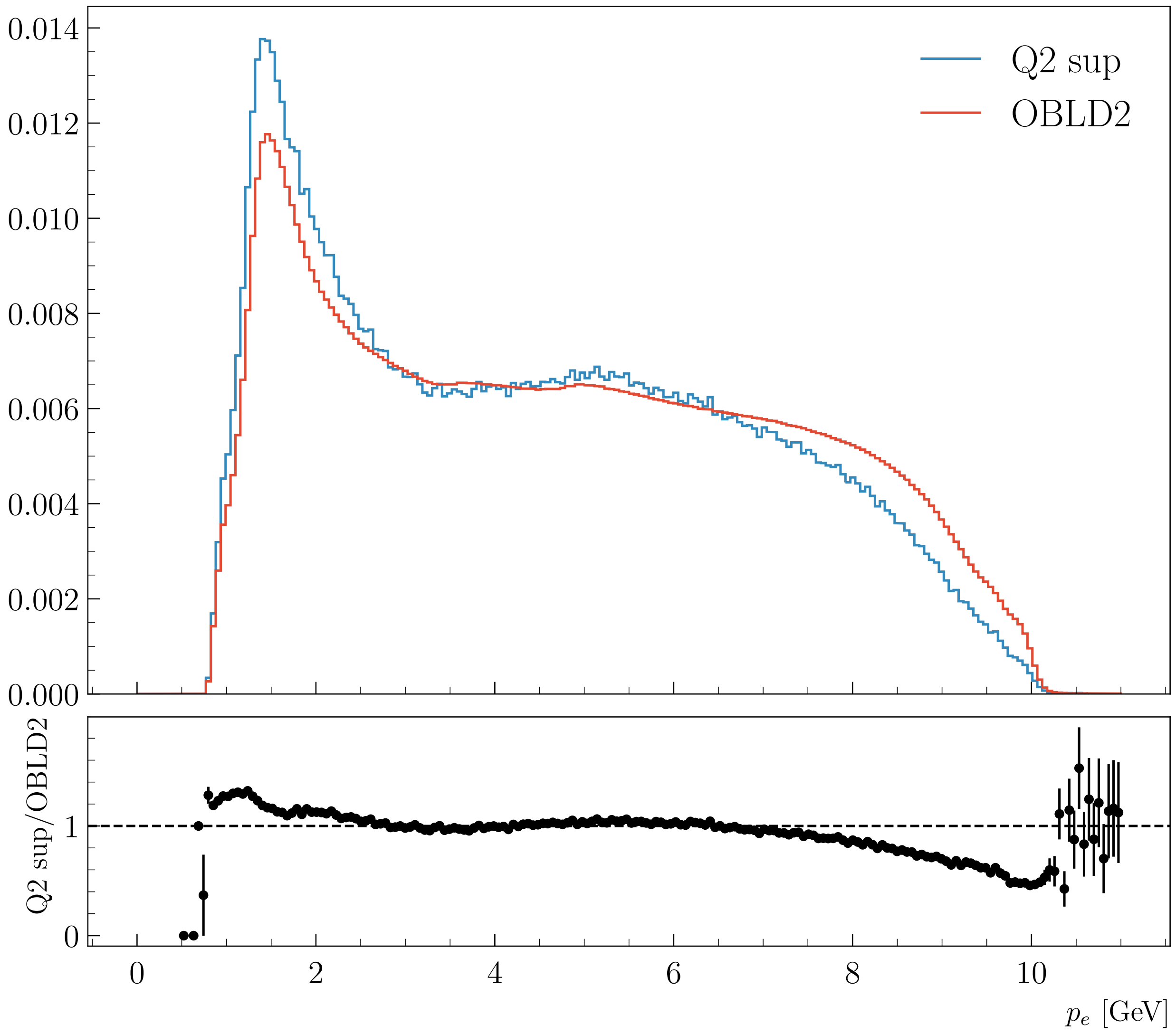


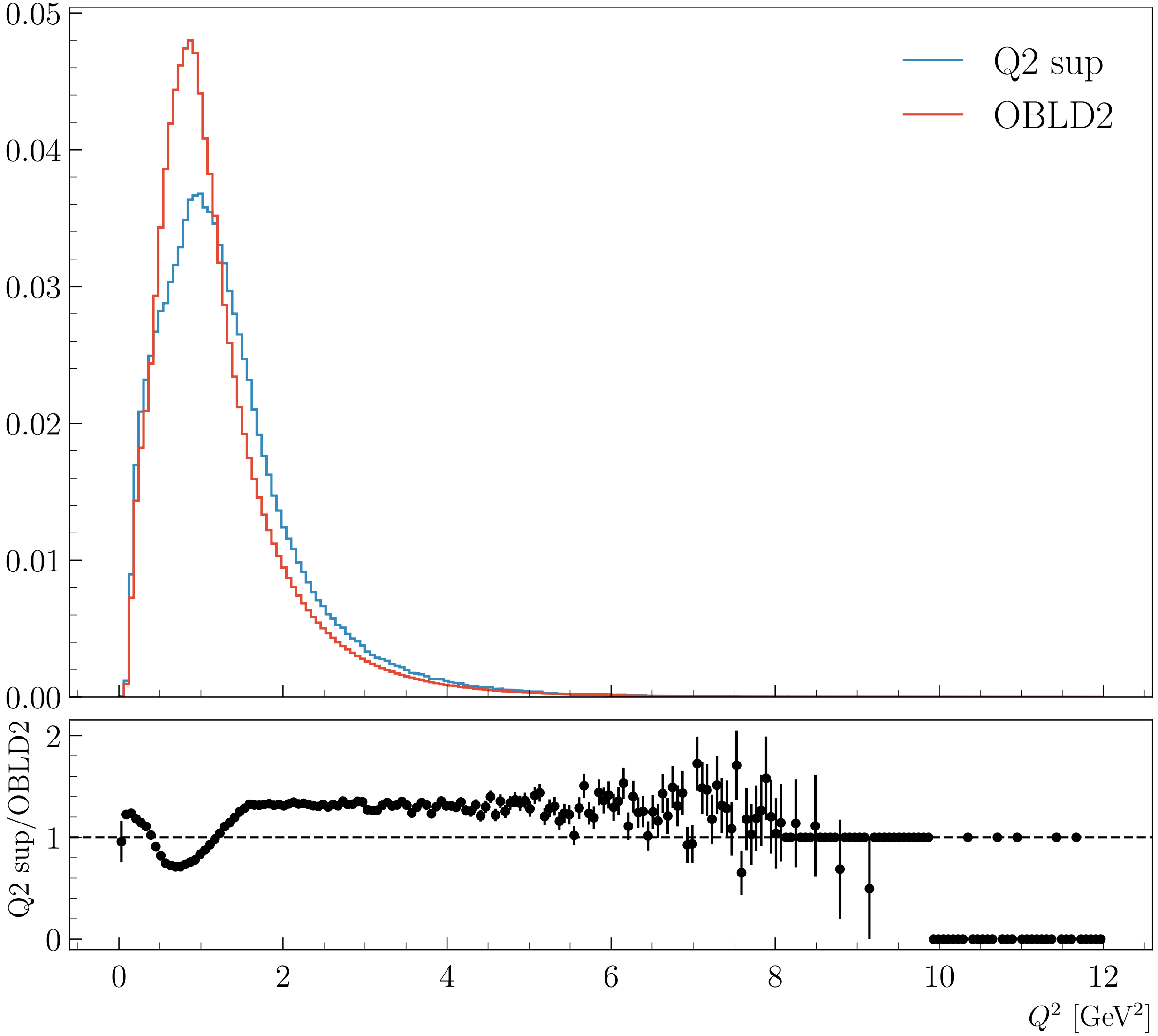
0.0200
0.0175
0.0150
0.0125
0.0100
0.0075
0.0050
0.0025
0.0000
Problematic DC
IBCuSn
5
0
Problematic DC/IBCuSn
0
25
50
75
100
125
150
175
θ_{π^-} [deg]

Physics quantities with problematics DC runs

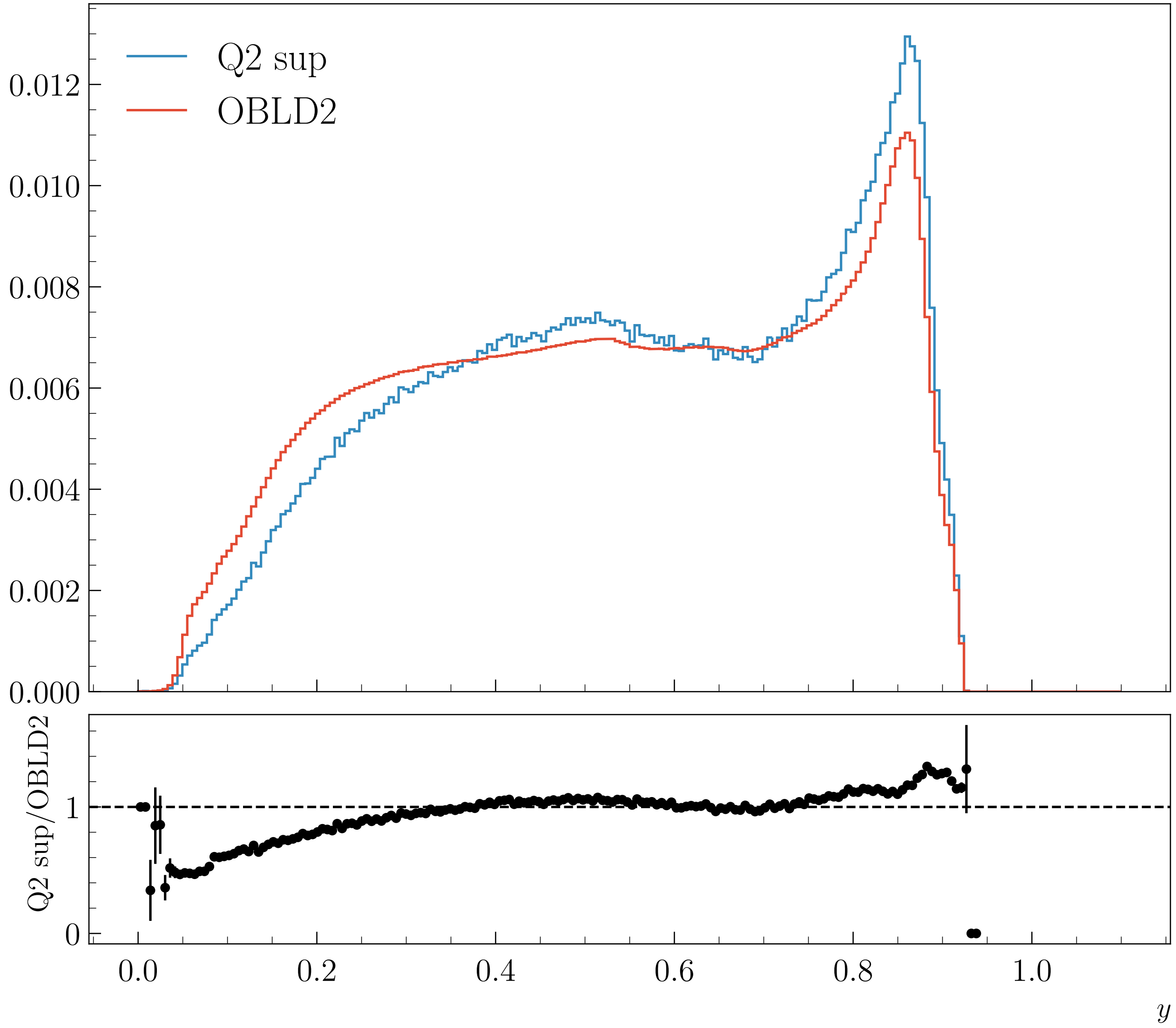


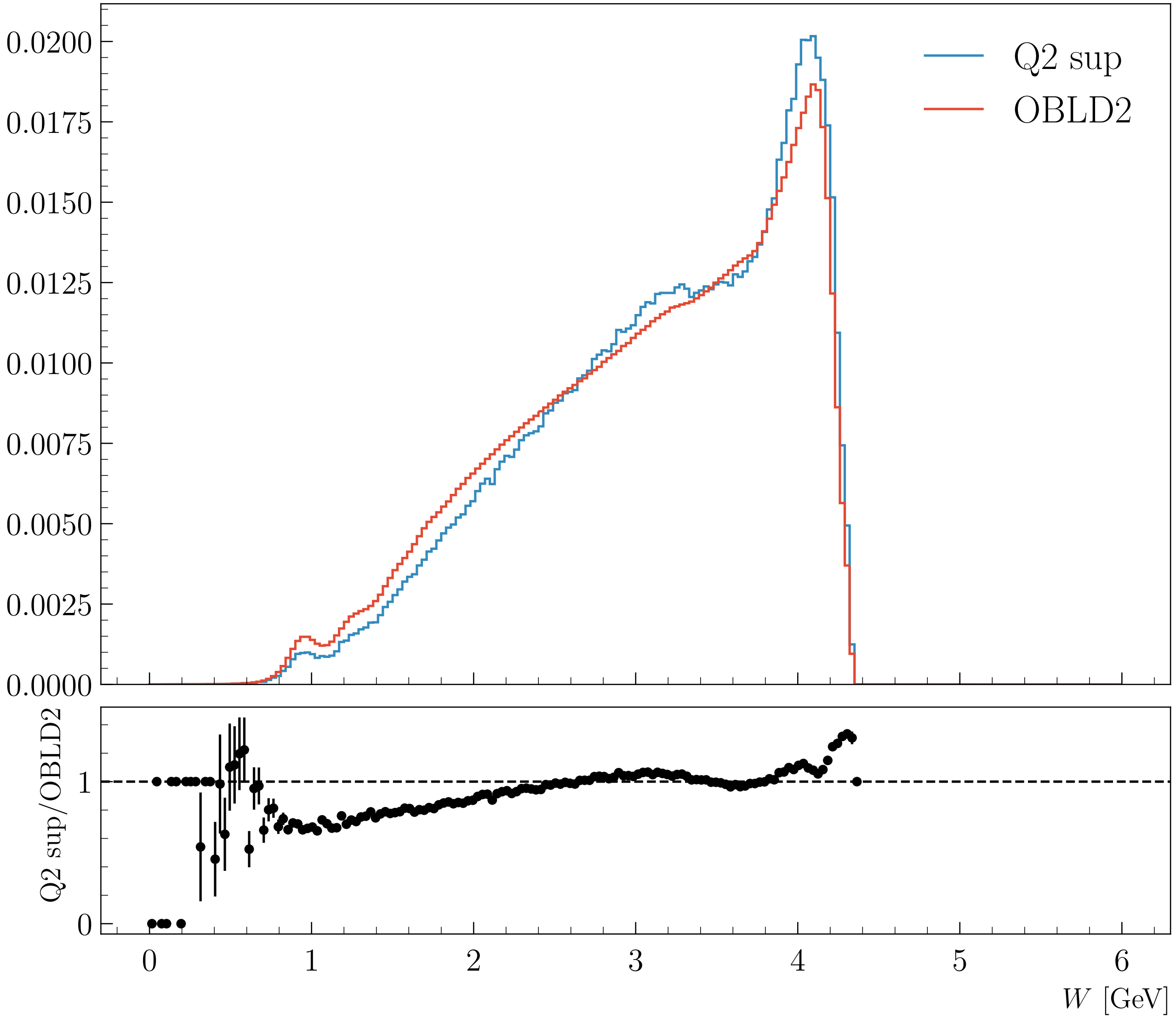
Physics quantities with trigger run 18432



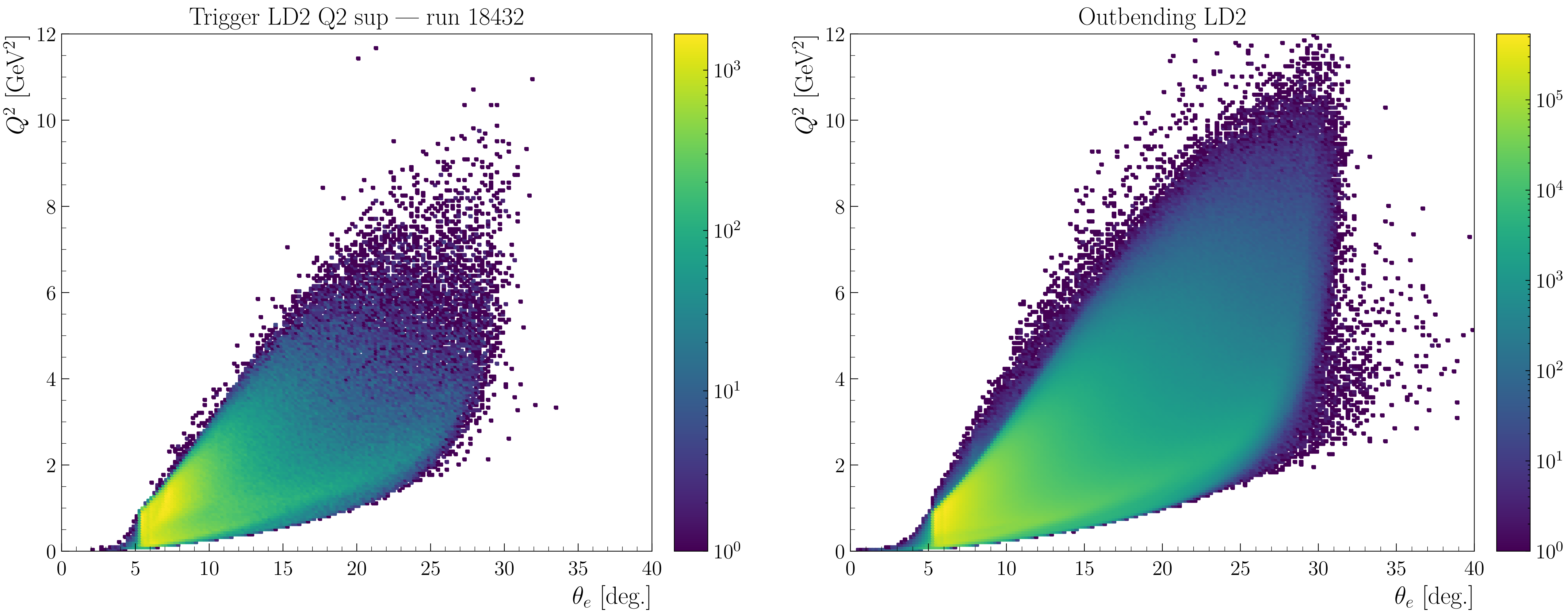


Physics quantities with trigger run 18432

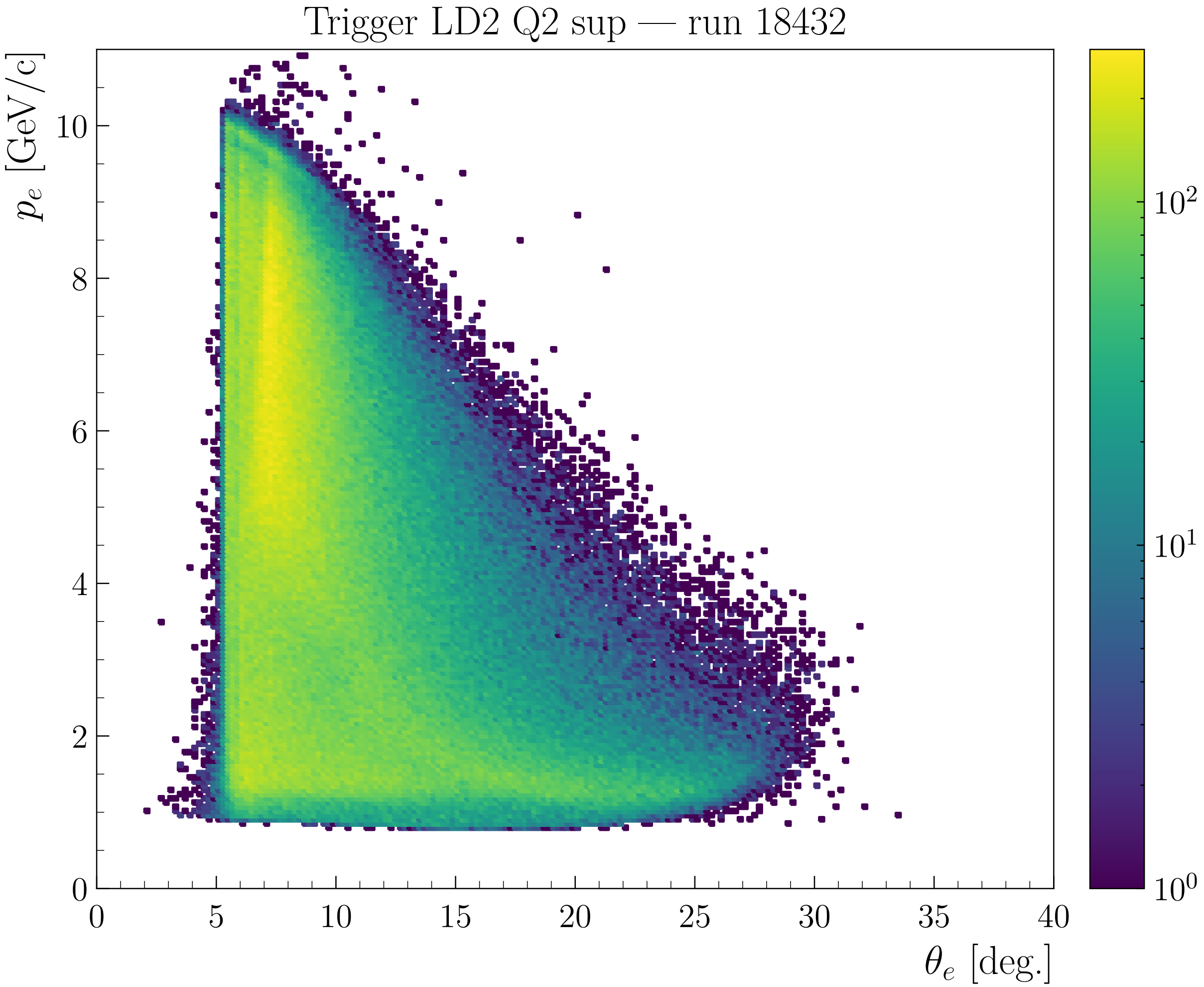


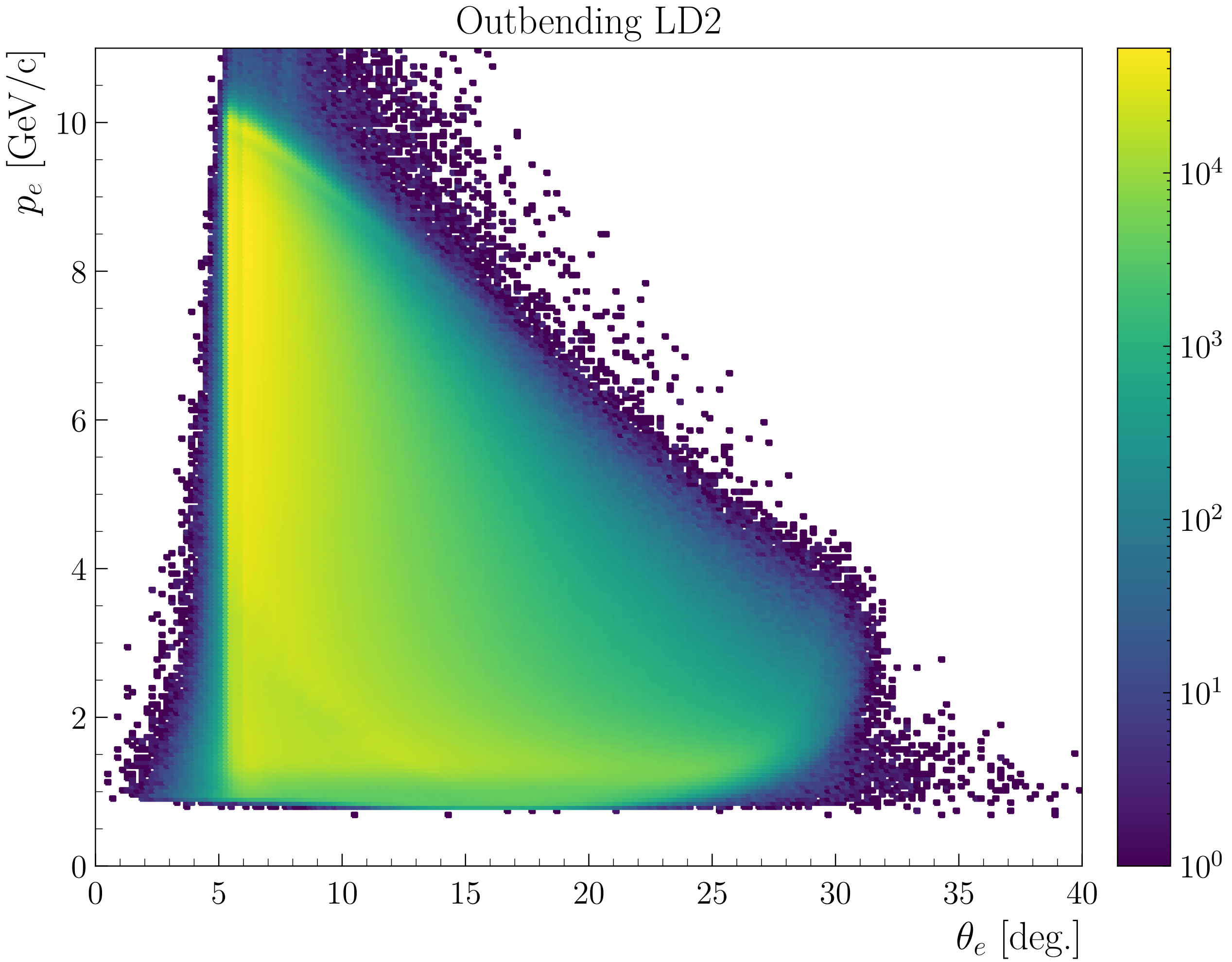


Physics quantities with trigger run 18432



Physics quantities with trigger run 18432





RG-D data size for pass1 review

Number/Data size

Number of files: 572,317

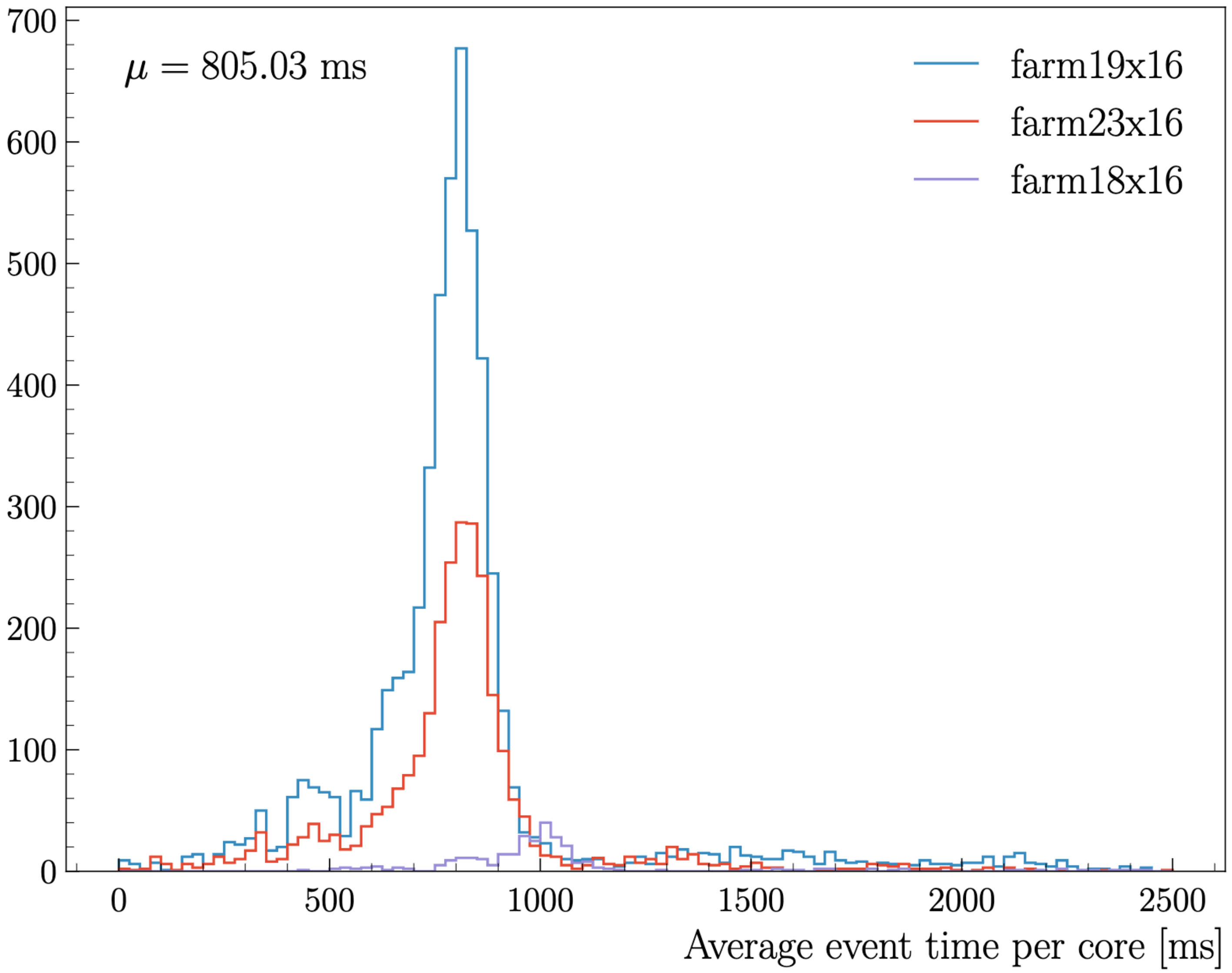
Number of runs: 612

Number of events: 39,715,110,220

Number of events per EVIO file: ~74,000

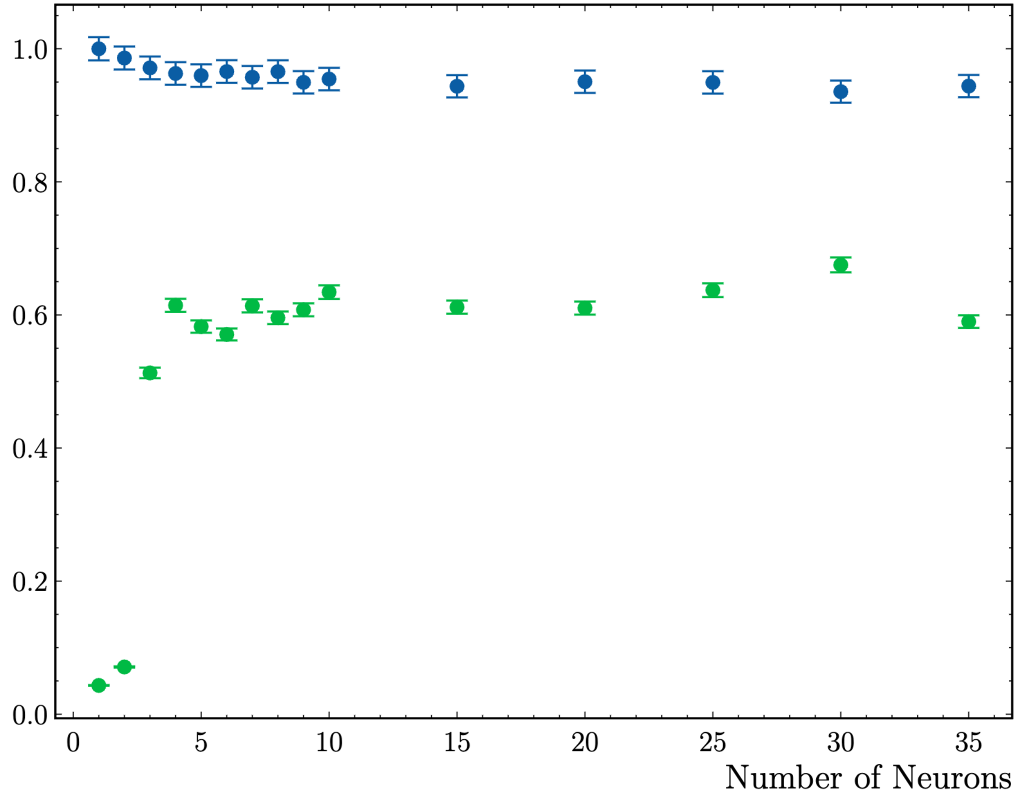
Size of raw file (2GB)	1144 TB on mss (2×572317)
Size of decoded files:	883T ($\frac{572317 \times 9.6\text{TB}}{6220}$)
Size of DST files:	199T (320G $\times$ 622 runs)
Size of DST skims files:	174T (Using my skim, should probably use a real skim (SIDIS))

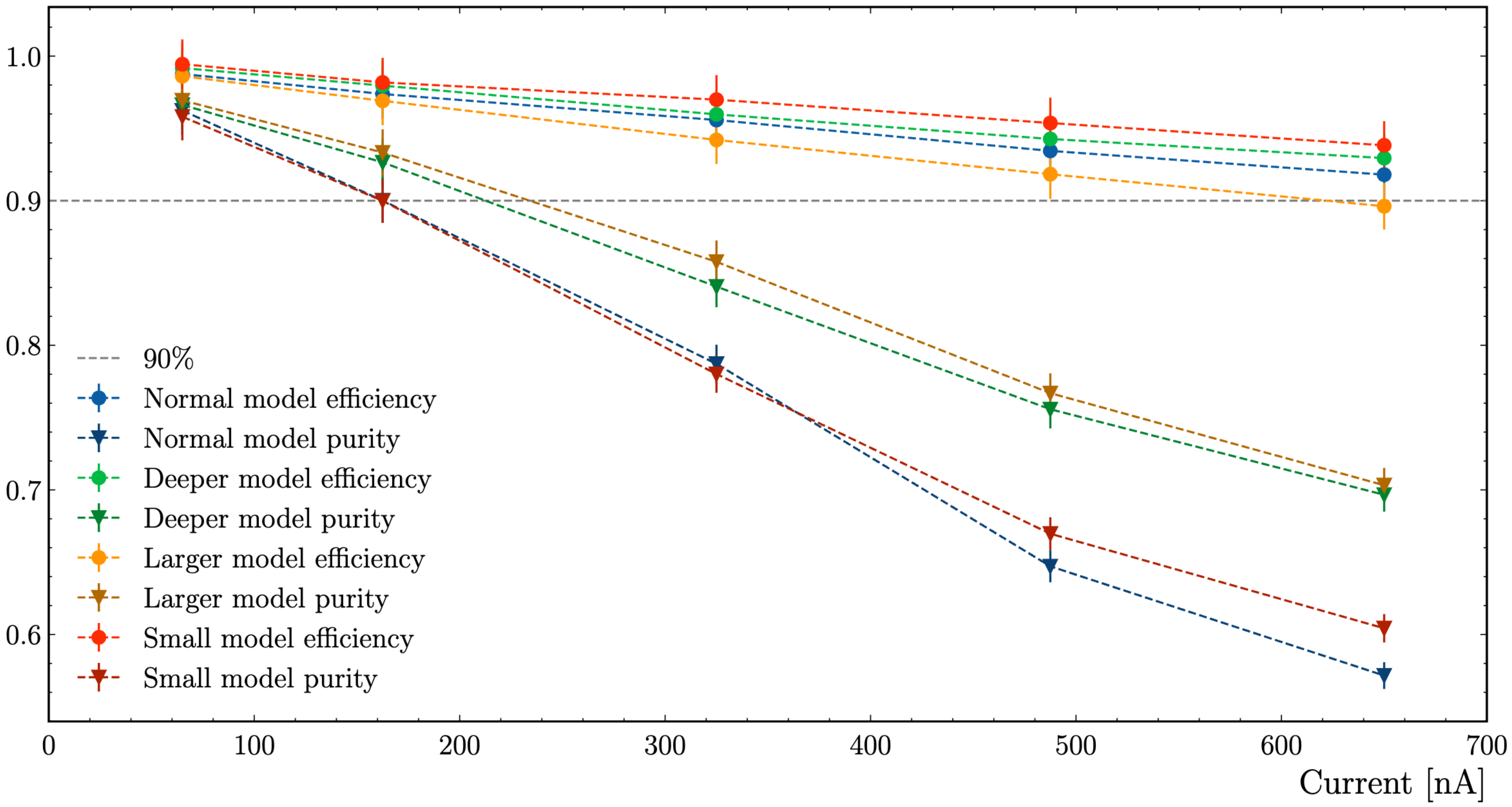
RG-D data size for pass1 review

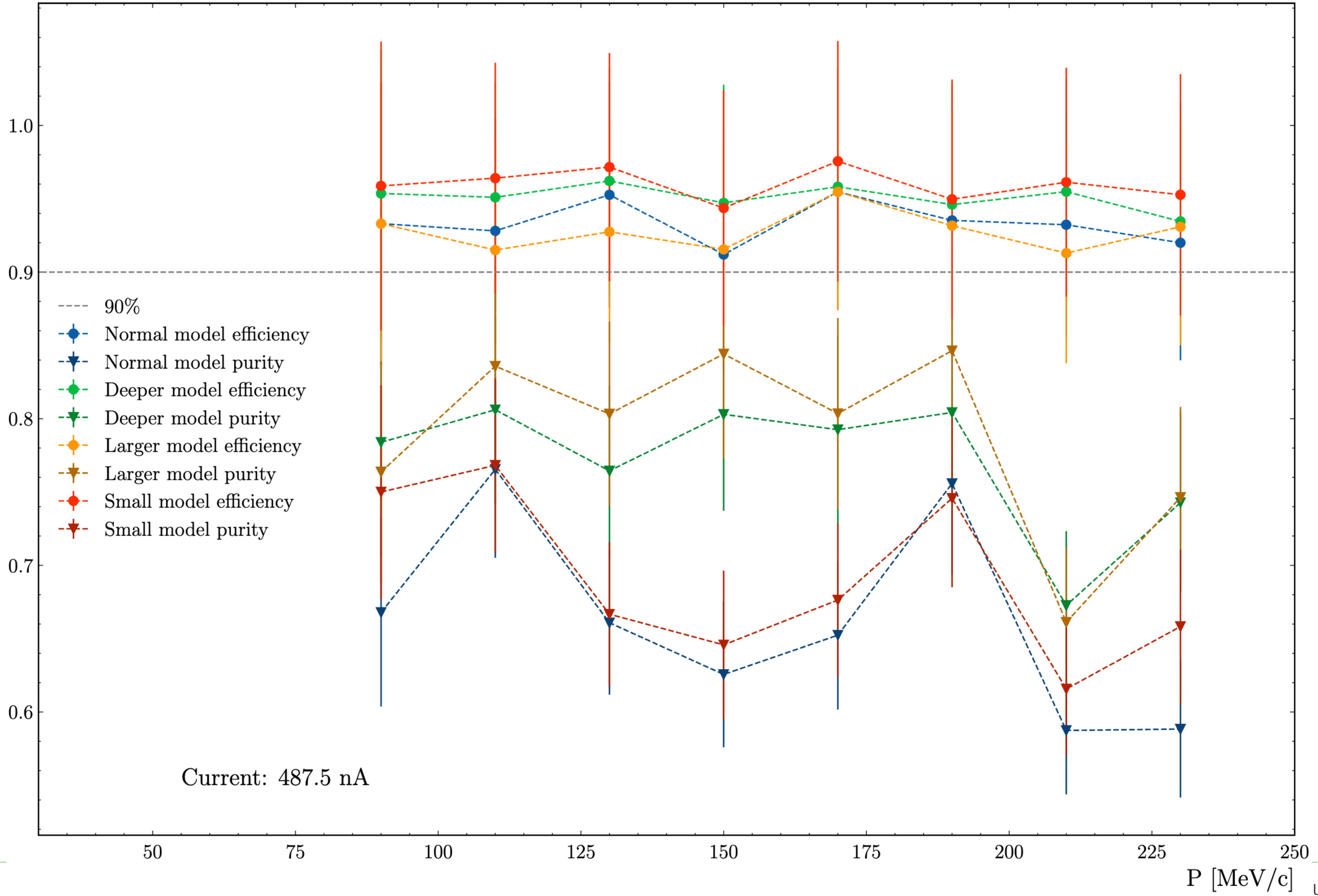


AI for ALERT

- The simulation is working now (12.0.4t).
- I have trained a new network with the new geometry
- I have tested networks with one layer with different number of neurons
- Blue: efficiency, green: purity
- Background added: 650nA
- Goal: To have good efficiency and purity.
- 1 neuron:
  - Efficiency: 1, but purity 0.05 -> not good
- 10 neurons:
  - Efficiency: 0.95, but purity 0.6 -> good




1.0
0.9
0.8
0.7
0.6
0
100
200
300
400
500
600
700
Current [nA]
90%
Normal model efficiency
Normal model purity
Deeper model efficiency
Deeper model purity
Larger model efficiency
Larger model purity
Small model efficiency
Small model purity


1.0
0.9
0.8
0.7
0.6
50
75
100
125
150
175
200
225
250
P [MeV/c]
90%
Normal model efficiency
Normal model purity
Deeper model efficiency
Deeper model purity
Larger model efficiency
Larger model purity
Small model efficiency
Small model purity
Current: 487.5 nA

To do list

Task	expected	done	comment
Redo the background merging studies	04/25	04/28	
Improve pid, fiducials cuts	04/21		Work on the DC fiducials, need to produce a PDF with all the plots and details
Produce physics plots for pass1 review	04/30		
Finish the event mixing method	05/04		
Extract the multiplicity ratio, pT	05/15		
improve code that generated all track candidates (ALERT)	05/15		
Test the AI on real data (ALERT)	04/21		Simulation work, new network have been trained, need to work on elastic data

One more thing

- Gabriel Charles organizes a workshop about gaseous detectors in France
- In June 18^{th} to 20^{th}
- He asked me if I could give a presentation about the AI-assisted track finding algorithm that has been developed for ALERT.
- The talk will be on Zoom, and there is no fee.